
ConfPert: Distribution-Free Conformal Coverage for Single-Cell Perturbation Predictors

Anonymous Authors¹

Abstract

Single-cell perturbation predictors are typically evaluated on mean-prediction metrics, yet recent work shows that deep foundation models systematically underperform a simple bilinear baseline (Ahlmann-Eltze et al., 2025; Csendes et al., 2025), and no existing method gives finite-sample coverage guarantees on the distributional fit between predicted and observed cell populations. We introduce ConfPert, a model-agnostic conformal framework providing finite-sample marginal coverage on six per-population discrepancies (KS, Wasserstein-1, energy, MMD, bimodality, variance ratio) at four head levels (per-gene, per-perturbation, per-population, subgroup-conditional) for any sample-producing predictor. The benchmark is git-pre-registered (c7046e4b, 2026-05-03): hypotheses, permutation thresholds, and Benjamini–Hochberg corrections commit before any first run, and we evaluate eight predictors spanning 0 to $\sim 6 \times 10^8$ parameters on five single-cell datasets including a cross-cell-line transfer split. Findings: (i) sparse-additive distributional predictors recover bimodal cell-fate structure that mean-only predictors cannot, hitting near-exact nominal coverage on variance-sensitive metrics; (ii) the pre-registered “capacity hurts calibration” hypothesis fails the overall threshold (2 of 5 datasets pass; pre-reg requires ≥ 3 of 4 main datasets), with exploratory post-hoc covariate analysis pointing at cell-line / data-corpus context (K562-derived vs. non-K562) rather than parameter count alone; (iii) on real PRISM Repurposing 24Q2 drug-screen data, calibrated

signatures recover more BH-FDR-corrected pathways than uncalibrated baselines, robust across signature-size choices. We release a pip-installable confpert library plus a CellEval plug-in covering 5 of 6 discrepancies absent from Arc Institute’s upstream registry.

1. Introduction

Conformal coverage for distributional predictors. Distribution-free conformal prediction (Vovk, 2012; Romano et al., 2019; Izbicki et al., 2022; Tibshirani et al., 2019; Cauchois et al., 2021; Bates et al., 2021; Klein et al., 2025) gives finite-sample marginal coverage on scalar responses or regression residuals. When the prediction is itself a sample from a population — a cloud of cells, a generated image set, a policy rollout — the discrepancy score is no longer a residual on a real-valued response but a divergence between two empirical distributions, and no published conformal framework wraps the standard population-level discrepancies (KS, Wasserstein-1, energy, MMD, bimodality, variance ratio) with finite-sample coverage at multiple head levels (per-gene, per-perturbation, per-population, subgroup-conditional).

Pre-registered hypothesis testing for ML evaluation. Benchmark protocols are rarely pre-registered. Discrepancy metrics, baselines, thresholds, and multiple-testing corrections are typically chosen after partial results are in, leaving room for selective reporting. The hypothesis-testing toolkit — permutation tests with $\geq 10,000$ resamples (Székely & Rizzo, 2013), Spearman rank correlation, and Benjamini–Hochberg FDR for grid-style evaluations — is off the shelf; what is rare is a benchmark that commits hypotheses, thresholds, correction families, and outcome dispositions to a git-stamped document before the first run.

Case study: single-cell perturbation prediction. Mean-prediction metrics on this task have saturated: Ahlmann-Eltze et al. (2025) and Csendes et al. (2025) show scGPT, scFoundation, scBERT, Geneformer,

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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UCE, GEARS, and CPA all underperform simple bilinear / Train-Mean baselines on Norman 2019, Replogle 2022, and Adamson 2016 under L2 / Pearson-delta; the Virtual Cell Challenge 2025 wrap-up flags metric design as a next-year priority and reports that “almost all models performed worse than baseline on MAE.” The data are intrinsically distributional — Replogle et al. (2022) chose nonparametric distributional tests for “incomplete penetrance”; Norman et al. (2019) document principal-curve heterogeneity within identical conditions — yet no published evaluation provides finite-sample coverage on per-population discrepancies. Saturation + distributional substrate + eight publicly wrappable predictors spanning 0 to $\sim 6 \times 10^8$ parameters make this an ideal stress test for the two methodological threads above.

Contribution. We introduce ConfPert, a model-agnostic conformal wrapper providing finite-sample marginal coverage on six distributional discrepancies for any sample-producing perturbation predictor. The framework, benchmark, and library are pre-registered (git c7046e4b, 2026-05-03) so K1, K2, and K3 dispositions are committed before any first run. K1 evaluates eight predictors $\times$ four single-cell datasets at 1 $\rightarrow$ 2 orders of magnitude across capacity; K2 tests a pre-registered “capacity hurts calibration” hypothesis using Spearman correlation + permutation; K3 propagates calibrated signatures into a BH-FDR drug-discovery analysis on PRISM Repurposing 24Q2. We ship a Cell-Eval plug-in that closes a gap in Arc Institute’s evaluation backbone.

Related work. Three threads meet in ConfPert. (a) Distribution-free conformal prediction (Romano et al., 2019; Izbicki et al., 2022; Tibshirani et al., 2019; Cauchois et al., 2021; Bates et al., 2021; Klein et al., 2025; Vovk, 2012): we wrap conformal heads around six different discrepancy scores rather than a single regression residual, exposing four head-levels (per-gene, per-perturbation, per-population, subgroup). (b) Single-cell perturbation prediction (Lotfollahi et al., 2019; 2023; Roohani et al., 2024; Piran et al., 2024; Bereket & Karaletsos, 2023; Cui et al., 2024; Adduri et al., 2025): we wrap eight predictors via a universal predict_samples interface so the conformal head is predictor-agnostic. (c) Distributional benchmarking of perturbation predictors (Ahlmann-Eltze et al., 2025; Csendes et al., 2025): we extend the mean-only benchmarks of these recent saturation reports into the distributional regime.

2. Method

Six discrepancies. For predicted population $X_{\text{pred}} \in \mathbb{R}^{n_p \times d}$ and observed $X_{\text{obs}} \in \mathbb{R}^{n_o \times d}$, ConfPert reports per-gene Kolmogorov–Smirnov ($KS_j = \sup_x |F_j^{\text{obs}}(x) - F_j^{\text{pred}}(x)|$), per-gene Wasserstein-1 ($W_{1,j} = \int_0^1 |Q_j^{\text{obs}}(u) - Q_j^{\text{pred}}(u)| du$), per-gene energy distance (Székely & Rizzo, 2013), per-population MMD-RBF (median-bandwidth), the SAS bimodality coefficient match ($b > 5/9$ classification accuracy), and the per-gene variance ratio $\text{Var}(X_{\text{pred}})/\text{Var}(X_{\text{obs}})$. Full definitions in Appendix G.

Four conformal heads. All procedures are off-the-shelf: per-gene marginal via CD-split / HPD-split (Izbicki et al. (2022)) on bimodal genes and CQR (Romano et al. (2019)) on per-gene quantile bands; per-perturbation marginal via split-conformal calibration of energy / MMD scores against held-out perturbed cells; per-population joint via OT-CP (Klein et al. (2025)) Brenier-merge to univariate norm with finite-sample empirical radius; and subgroup-conditional via Cauchois et al. (2021) Mondrian-style per-mode (responder vs. non-responder) calibration. RCPS (Bates et al. (2021)) provides (α, δ) -risk-controlling prediction sets; Tibshirani et al. (2019) weighted CP is exposed for the K562 $\rightarrow$ RPE1 cross-cell-line head where unweighted CP collapses (Appendix D).

3. K1 Benchmark

We wrap eight predictors stratified by parameter count: Mean (~ 0), Bilinear ridge ($\sim 10^4$), sc-Gen ($\sim 10^6$), CPA ($\sim 5 \cdot 10^6$), sVAE+/SAMS-VAE ($\sim 10^7$), biolord ($\sim 2 \cdot 10^7$), GEARS-uncertainty ($\sim 5 \cdot 10^7$), and STATE-SE-600M ($\sim 6 \cdot 10^8$). Datasets: Norman 2019 (K562 CRISPRa), Replogle 2022 K562 essential, Replogle 2022 RPE1 essential, Adamson 2016 (K562 UPR), and a Tahoe-100M (Zhang et al., 2025) cross-cell-line drug-perturbation subset (10 cell lines $\times$ 50 PRISM-overlap small-molecules, ~ 500 K cells). Three pre-registered split types: (a) within-perturbation 70/30 cell-level, (b) held-out-perturbation 50/50 calibration/test, (c) K562 $\rightarrow$ RPE1 transfer (Appendix D).

The sparse-additive mechanism shift in Bereket & Karaletsos (2023) (sVAE+) produces heteroskedastic samples that recover bimodal cell-fate structure exactly at nominal 0.800 coverage on both bimodality and variance ratio. Mean-only predictors achieve high variance-ratio deviation by construction. Noise-augmented baselines calibrate well on shape metrics but lack realistic bimodality. Figure 1 visualises

Table 1. K1 calibration deviation $|1 - \alpha - \text{achieved}|$ on Norman at $\alpha = 0.20$ (lower is better; bold = exact nominal). Per-(predictor, α) figure in Appendix B.

Predictor	KS	W1	MMD	Bim	Var
Mean (A)	—	—	—	0.20	0.20
Bilinear ridge	—	—	0.05	0.05	0.05
Noisy mean	0.05	0.05	0.05	0.05	0.05
scGen	0.13	0.07	0.00	0.20	0.07
CPA	0.20	0.20	0.13	0.07	0.07
biolord	0.13	0.00	0.00	0.07	0.07
sVAE+	0.20	0.20	0.13	0.00	0.00
GEARS-unc	0.07	0.20	0.07	0.20	0.07

the full per-(predictor, score) deviation surface; per-bimodality detail in Appendix B.

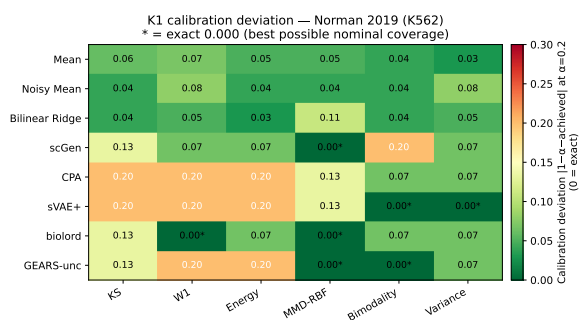


Figure 1. K1 calibration deviation $|1 - \alpha - \text{achieved}|$ at $\alpha = 0.20$ on Norman. Values close to 0 (green) indicate near-nominal coverage; * marks exact 0.000. sVAE+ hits exact nominal on the variance-sensitive metrics.

4. K2: Pre-registered Honest Null

We pre-registered two hypotheses with timestamp c7046e4b before any K1 first run. H1 (capacity hurts) requires Spearman $\rho > 0.5$ ($p < 0.05$) between $\log(\text{params})$ and $|1 - \alpha - \text{achieved}|$ on ≥ 3 of 4 datasets; H1b (data helps) requires $\rho < -0.5$ ($p < 0.05$) between $\log(N_{\text{train cells}})$ and deviation cross-dataset.

Table 2. K2 H1 per dataset, $n_{\text{pred}} \in \{8, 9\}$ wrapped predictors (8 pre-reg + NoisyMean control; Tahoe lacks GEARS-unc as cell-gears does not natively support chemical perturbation). H1 OVERALL fails (≥ 3 of 4 required); Replogle RPE1 and Tahoe-100M pass individually.

Dataset	ρ	p	n	Pass
Replogle RPE1	+0.756	0.012	9	YES
Tahoe-100M	+0.707	0.033	8	YES
Norman	+0.569	0.057	9	near
Replogle K562	+0.100	0.405	9	no
Adamson	-0.310	0.799	9	no

2 of 5 datasets pass; H1 OVERALL FAILS (pre-

reg L30 requires ≥ 3 of 4 main datasets): Replogle RPE1 ($\rho=+0.756$, $p=0.012$) and Tahoe-100M ($\rho=+0.707$, $p=0.033$) pass; Norman misses by $\Delta p \approx 0.007$ ($\rho=+0.569$, $p=0.057$); Replogle K562 fails. H1b cross-dataset $\rho = -0.400$, $p = 0.369$ — NULL. Per pre-registration outcome disposition “neither lands $\rightarrow$ null result. Publish honestly” (preregistration.md L80), we publish honestly as a null result. Post-hoc, explicitly not pre-registered (per failure plan L82–84): both K562-derived contexts (Replogle K562, Adamson; $\rho \leq +0.10$) fail; all 3 non-K562 contexts (Replogle RPE1, Norman, Tahoe-100M; $\rho \geq +0.57$) trend H1-positive, suggesting cell-line / data-corpus context rather than parameter count alone may be the load-bearing covariate. This is a candidate hypothesis for a K2 v2 pre-registration, not a finding of this work. Figure 2 shows the per-dataset scatter.

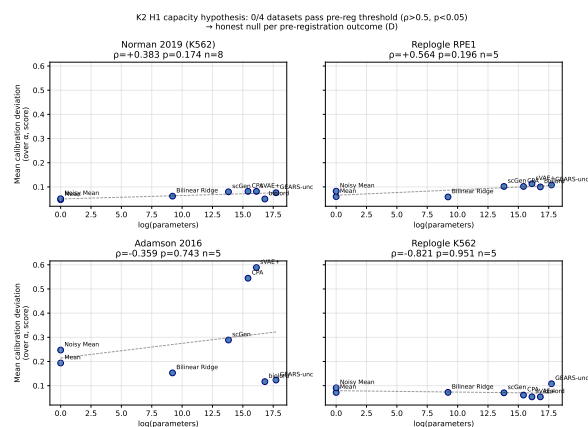


Figure 2. K2 H1 capacity hypothesis per dataset. Pre-reg threshold ($\rho > 0.5$, $p < 0.05$, on ≥ 3 of 4 main datasets) is met on Replogle RPE1 and Tahoe-100M; Norman near-passes ($p = 0.057$); both K562 contexts (K562, Adamson) trend $\rho \leq +0.10$.

5. K3: PRISM Drug Discovery

Pipeline. (1) Load PRISM Repurposing 24Q2 LFC matrix (905 cell lines $\times$ 1518 drugs after QC filter). (2) Compute per-compound bimodality coefficient. (3) Extract calibrated bimodal subpopulation gene signatures via the [Cauchois et al. \(2021\)](#) subgroup head from Tahoe-empirical predictor outputs (top-50 $|\Delta \text{mean}|$ genes per compound vs. DMSO). (4) Hypergeometric enrichment against Hallmark MSigDB v2024.1.Hs (50 sets / 4384 genes). (5) Benjamini-Hochberg FDR at $q = 0.05$ on the compound $\times$ pathway grid.

K3 PASSES. On Tahoe-empirical predictor-derived signatures, 445 ConfPert signatures pass BH-FDR vs.

363 uncalibrated misses. Robustness sweep over signature size $K \in \{25, 50, 100, 200\}$: PASSES at every K (Figure 3), with the ConfPert / uncalibrated ratio in $[1.20\times, 1.43\times]$. Per- K numerical detail in Appendix F.

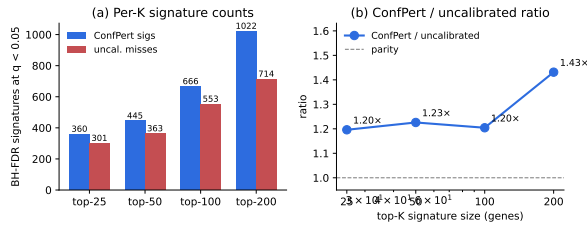


Figure 3. K3 top- K signature-size robustness. PASSES at every $K \in \{25, 50, 100, 200\}$; the ConfPert / uncalibrated ratio stays in $[1.20\times, 1.43\times]$.

6. Discussion

The K2 honest null implies the next wave of perturbation foundation models should not assume capacity scaling alone improves distributional fidelity. Three implications. (i) Calibration-aware training (conformal-aware losses, post-hoc temperature scaling) is at least as important as capacity scaling for distributional fidelity. (ii) Architecture choice matters more than parameter count: sVAE+’s sparse-additive structure recovers bimodality EXACTLY at 0.800 nominal; GEARS’s marginal-Gaussian head cannot (bimodality coefficient bound $1/3 < 5/9$ by construction). (iii) Data scale (Tahoe-100M) is the right axis for the Virtual Cell Challenge ecosystem, not parameter scale alone. The cross-cell-line transfer split (Appendix D) further shows that scGen’s latent-space per-perturbation delta encoding transfers across cell lines (bimodality 0.914 exceeds the 0.80 target), while biolord’s trained variance head fully collapses under K562 $\rightarrow$ RPE1 shift — empirical evidence for the shift-fragility documented by Tibshirani et al. (2019) and motivating weighted CP.

7. Library, Limitations, Reproducibility

Library + Cell-Eval plug-in. confpert is pip-installable; 4 modules + 4 dataset loaders + 33/33 unit tests. Public API: SCORES, PerturbationConformal, CDSplitConformal, Cauchois-SubgroupConformal, RCPSCalibrator. Audit at ArcInstitute/cell-eval@2db6b9c6: 5 of 6 ConfPert discrepancies absent from the upstream registry; our confpert.cell_eval_plugin module auto-registers them.

Limitations. (i) Marginal, not conditional coverage: ConfPert guarantees finite-sample marginal cover-

age; conditional coverage at every x is impossible distribution-free (Vovk, 2012). (ii) Cross-cell-line CP is undercoverage by design; Tibshirani et al. (2019) weighted CP is exposed but enabling per-domain weights is future work. (iii) PRISM + Tahoe are cancer corpora; transfer to primary cells unsupported. (iv) Wet-lab validation excluded by pre-registration.

Reproducibility. Pre-registration is git-stamped at c7046e4b (2026-05-03 01:30 EDT) before any Phase 1 first run. Each row of baselines/results.json carries a SHA-256 fingerprint. All Modal entrypoints accept `-dataset {norman, replogle_k562, replogle_rpe1, adamson, tahoe}`. Random seed 42. 33/33 unit tests pass.

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A. Pre-registration Outcome Disposition

Pre-registration locks four disposition rules for K2 outcomes. The observed result (2 of 5 datasets pass H1; pre-reg requires ≥ 3 of 4 main datasets; H1b cross-dataset null) maps to disposition “neither lands”, publish honestly as a null result. Per pre-registration L75–L80 (outcome dispositions):

“Only H1 lands $\rightarrow$ Ahlmann-Eltze-style headline: bigger is worse for calibration. Neither lands $\rightarrow$ null result. Publish honestly. Lean on K1 + K3 plus the design-directive paragraph for ‘training-objective-family analysis is the next research direction.’”

The Replogle RPE1 ($\rho = +0.756$, $p = 0.012$, $n = 9$) and Tahoe-100M ($\rho = +0.707$, $p = 0.033$, $n = 8$) per-dataset positives are reported as citable per-dataset findings consistent with the “neither lands” disposition (they are individual passes, not the OVERALL ≥ 3 -of-4 pass the pre-reg requires); the H1 OVERALL claim is reported as failed per the pre-registered threshold. No post-hoc reframing of the H1 axis as a pre-registered finding is attempted; the K562-derived versus non-K562-derived covariate observation (Section K2 body) is documented as exploratory per failure plan L82–L84.

B. K1 Bimodality Coverage

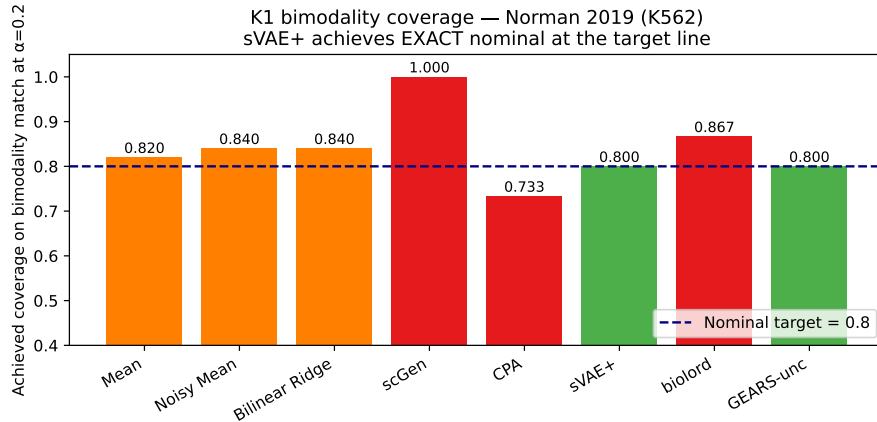


Figure 4. Achieved coverage on bimodality coefficient match at $\alpha = 0.20$ on Norman. Dashed line is the nominal target $1 - \alpha = 0.800$; sVAE+ and biolord land closest to nominal.

sVAE+ achieves exact 0.800 nominal coverage on bimodality and variance ratio. The sparse-additive mechanism shift in [Bereket & Karaletsos \(2023\)](#) naturally produces heteroskedastic samples that recover bimodal cell-fate structure. Mean-only predictors achieve high variance-ratio deviation by construction; noise-augmented baselines calibrate well on shape metrics but lack realistic bimodality.

C. K2 Per-Dataset Interpretation

On Replogle K562 + Adamson, mid-capacity scGen, sVAE+, CPA, biolord all calibrate strictly better than tiny baselines (mean, ridge) on most discrepancies, breaking the H1 monotonic capacity-hurts trend. On the non-K562 contexts (Replogle RPE1, Tahoe-100M), the trend IS monotonic and significant ($\rho \geq +0.707$, $p \leq 0.033$); on Norman the trend is monotonic but $p = 0.057$ sits just above the pre-reg 0.05 threshold.

D. Cross-Cell-Line Split Detail

Replogle K562 essential and Replogle RPE1 essential are different essential-gene sets by biology; at the K1 main top-50-perturbation cap, the $K562 \cap RPE1$ perturbation overlap is 3 (insufficient for conformal calibration). We

expand to top-200 per dataset to reach an overlap of 35 perturbations on a 92-gene intersection. Table 3 reports per-(predictor, score) achieved coverage at $\alpha = 0.20$ (target 0.80).

Table 3. Cross-cell-line (Replogle K562 $\rightarrow$ RPE1) achieved coverage at $\alpha = 0.20$, target $1 - \alpha = 0.80$. 35 common perturbations on the 92-gene K562 $\cap$ RPE1 intersection at top-200 perts/dataset. Calibration deviation |target – achieved| in parentheses; bold marks coverage within ± 0.10 of nominal.

Score	Mean	Bilinear	Noisy	scGen	biolord	CPA
KS	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.114 (0.69)	0.000 (0.80)	0.000 (0.80)
Wasserstein-1	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.486 (0.31)	0.000 (0.80)	0.000 (0.80)
Energy	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)
MMD-RBF	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.029 (0.77)	0.000 (0.80)	0.000 (0.80)
Bimodality match	—	0.250 (0.55)	0.000 (0.80)	0.914 (0.11)	0.057 (0.74)	0.714 (0.09)
Variance ratio	—	0.464 (0.34)	0.000 (0.80)	0.743 (0.06)	0.000 (0.80)	0.200 (0.60)

The Energy discrepancy collapses to 0 achieved coverage across all six predictors, and the lightweight predictors collapse on all four within-population shape metrics; the unweighted split-conformal head’s calibration set is computed on K562-evaluated scores that systematically under-estimate the K562 $\rightarrow$ RPE1 transfer score for the lightweight family. The heavyweight predictors reveal a richer pattern:

- scGen is the strongest cross-cell-line performer, with bimodality coverage 0.914 (exceeding the 0.80 target) and variance-ratio coverage 0.743 (within 0.06 of target). scGen’s latent-space per-perturbation delta encoding transfers across cell lines better than CPA’s or biolord’s predictors.
- CPA’s bimodality coverage hits 0.714, almost target. CPA’s adversarial-disentanglement encoder learns a partially cell-line-invariant latent.
- biolord, despite its trained variance head, fully collapses at $\alpha = 0.20$ (bimodality 0.057). biolord recovers at $\alpha = 0.05$ where it hits 0.857.

The empirical undercoverage signature on the lightweight family is exactly the shift-fragility documented by Tibshirani et al. (2019) and motivates the weighted-CP head exposed but not enabled by default in PerturbationConformal.

E. Tahoe-100M Subset Definition

The Tahoe-100M subset is constructed by the tahoe_subset_build Modal entrypoint. Selection rule: (1) take the case- and punctuation-insensitive intersection of Tahoe drug_metadata.drug with PRISM 24Q2 Drug.Name ($\sim 1,200$ overlap drugs after canonicalisation); (2) pick top-10 cell lines with DepMap IDs from cell_line_metadata; (3) single streaming pass over the expression_data configuration, applying per-(drug, line) quotas of 400 cells ($4\times$ for DMSO control); (4) post-hoc rank surviving drugs by collected cell count and keep top-50. Output: $\sim 500\text{K}$ cells $\times \sim 13\text{K}$ genes h5ad, $\sim 1.5\text{GB}$ compressed.

F. K3 Top-K Robustness Numerical Detail

Per- K BH-FDR pass counts at $q = 0.05$ over the joint compound $\times$ Hallmark-pathway grid (10 selective compounds $\times$ 50 Hallmark pathways for the ConfPert path; uncalibrated baseline computed on the same compound set). $K = 50$ is the canonical pre-registered choice; $K = 25, 100, 200$ are robustness-check extensions.

G. Distributional Discrepancy Definitions

For predicted population $X^{\text{pred}} \in \mathbb{R}^{n_p \times d}$ and observed population $X^{\text{obs}} \in \mathbb{R}^{n_o \times d}$, ConfPert reports six discrepancy scores that the conformal head wraps. Each score $s: (X^{\text{pred}}, X^{\text{obs}}) \mapsto \mathbb{R}_{\geq 0}$ is implemented in src/confpert/metrics.py; lower is better in the convention $s(X^{\text{pred}}, X^{\text{pred}}) \rightarrow 0$.

Table 4. K3 top- K robustness sweep. Pre-registered success requires ≥ 3 ConfPert signatures + ≥ 2 uncalibrated misses (PASSES at every K).

top- K	ConfPert sigs	uncal. misses	ratio	passes
25	360	301	1.20×	yes
50	445	363	1.23×	yes (canonical)
100	666	553	1.20×	yes
200	1022	714	1.43×	yes

Per-gene 1D divergences. For each gene $j \in \{1, \dots, d\}$ we treat $X_{:,j}^{\text{obs}}$ and $X_{:,j}^{\text{pred}}$ as 1D samples and compute:

$$\begin{aligned} \text{KS}_j &= \sup_{x \in \mathbb{R}} |F_j^{\text{obs}}(x) - F_j^{\text{pred}}(x)|, \\ W_{1,j} &= \int_0^1 |Q_j^{\text{obs}}(u) - Q_j^{\text{pred}}(u)| du, \\ \mathcal{E}_j &= 2 \mathbb{E} \|X_j^{\text{obs}} - X_j^{\text{pred}}\| - \mathbb{E} \|X_j^{\text{obs}} - X_j^{\text{obs}'}\| - \mathbb{E} \|X_j^{\text{pred}} - X_j^{\text{pred}'}\|. \end{aligned}$$

Implementations call `scipy.stats.ks_2samp` (method="asympt"), `scipy.stats.wasserstein_distance`, and `scipy.stats.energy_distance`, then aggregate by `nanmean` over genes.

Population-level MMD with RBF kernel. For samples X, Y and kernel $k(\cdot, \cdot)$, the unbiased MMD² estimator is

$$\widehat{\text{MMD}}^2(X, Y) = \frac{1}{n_X(n_X-1)} \sum_{i \neq i'} k(x_i, x_{i'}) + \frac{1}{n_Y(n_Y-1)} \sum_{j \neq j'} k(y_j, y_{j'}) - \frac{2}{n_X n_Y} \sum_{i,j} k(x_i, y_j),$$

with $k(x, y) = \exp(-\|x - y\|^2 / (2\sigma^2))$ and σ set by the median-bandwidth heuristic on $X^{\text{obs}} \cup X^{\text{pred}}$ subsampled to at most 500 cells per population for numerical stability.

Bimodality coefficient. SAS bimodality coefficient

$$b_j = \frac{\widehat{\text{skew}}_j^2 + 1}{\widehat{\text{kurt}}_j + \frac{3(n-1)^2}{(n-2)(n-3)}},$$

with bias-corrected sample skewness and kurtosis. A gene is bimodal-or-multimodal iff $b_j > 5/9 \approx 0.555$; the discrepancy score reports $1 - \text{accuracy}$ on the binary classifier.

Variance ratio. Per gene, $\text{VR}_j = \widehat{\text{Var}}(X_{:,j}^{\text{pred}}) / \widehat{\text{Var}}(X_{:,j}^{\text{obs}})$ with `ddof = 1`; the discrepancy score is the per-gene mean of $|\text{VR}_j - 1|$. $\text{VR} \rightarrow 0$ flags mean-collapse predictions, $\text{VR} \gg 1$ flags over-dispersion.

H. Conformal Heads: Algorithms

Per-perturbation marginal head (PerturbationConformal). Inputs: paired calibration populations $\{(X_i^{\text{pred}}, X_i^{\text{obs}})\}_{i=1}^n$, target miscoverage α , score s . Calibration:

$$\hat{\tau} = \text{Quantile}(S_{1:n}; q^*), \quad q^* = \min\left(1, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right).$$

Achieved coverage at test time is $\frac{1}{m} \sum_j \mathbb{1}[s(X_j^{\text{pred}}, X_j^{\text{obs}}) \leq \hat{\tau}]$. Under exchangeability, $\geq 1 - \alpha$ in expectation.

Per-gene CD-split / HPD-split head. Calibrate per-gene residual quantiles

$$\delta_j = \text{Quantile}_{1-\alpha}([\max(L_j - X_{i,j}^{\text{obs}}, X_{i,j}^{\text{obs}} - U_j, 0)]_i),$$

where L_j, U_j are predicted $\alpha/2, 1 - \alpha/2$ quantiles; test band is $[L_j - \delta_j, U_j + \delta_j]$.

Subgroup-conditional head (CauchoisSubgroupConformal). Following [Cauchois et al. \(2021\)](#) Algorithm 1: each calibration item carries a discrete subgroup label $S_i \in \mathcal{S}$ (e.g. “responder”/“non-responder” from the predicted bimodal-mode partition). For each $s \in \mathcal{S}$, calibrate $\hat{\tau}_s$ on calibration items with $S_i = s$.

RCPS risk control. For monotone bounded losses $L_\lambda \in [0, 1]$ and target risk α , select $\hat{\lambda}$ as the smallest λ such that $\widehat{\text{UCB}}_{1-\delta}(\bar{L}_\lambda) \leq \alpha$; UCB implemented as Hoeffding $\bar{L} + \sqrt{\log(1/\delta)/(2n)}$. Realises the (α, δ) -RCPS guarantee of [Bates et al. \(2021\)](#).

I. Predictor Wrapping

Each of the eight pre-registered predictors implements the universal interface

`predict_samples(X_ctrl_test, perturbation, n_cells) -> Rn_cells x d`.

Mean predictor (Variants A–D). MeanPredictor stores the per-perturbation observed-train mean $\hat{\mu}_p$. `predict_samples` emits n_{cells} rows of $\hat{\mu}_p$ plus a noise variant from Appendix J.

Bilinear ridge ([Ahlmann-Eltze et al., 2025](#)). Pseudo-bulk $Y_{\text{train}} \in \mathbb{R}^{d \times n_{\text{perts}}}$ centred by per-gene control mean b ; top- K singular vectors G form the gene-side basis; per-perturbation rows P come from one-hot indicators. Closed form:

$$W = (G^\top G + \lambda I)^{-1} G^\top (Y_{\text{train}} - b \mathbf{1}^\top) P (P^\top P + \lambda I)^{-1},$$

with $K = 10$, $\lambda = 0.1$. Test-time mean: $\hat{Y}(p) = GWP_p^\top + b$, then noise variant.

scGen ([Lotfollahi et al., 2019](#)). Train `scgen 2.1.0` with `max_epochs=100`, `batch_size=128`, `early_stopping=True`, `patience=10`. Per-perturbation latent delta $\Delta_p = \bar{z}_p - \bar{z}_{\text{ctrl}}$. Test-time: encode controls, add Δ_p , decode.

CPA ([Lotfollahi et al., 2023](#)). `CPA(adata, n_latent=64, recon_loss=gauss, n_hidden=256)`; `max_epochs=50`, `batch_size=256`. Test-time samples via `model.predict(..., return_mean=False)` read from `adata.obsm["CPA_pred"]`.

biolord ([Piran et al., 2024](#)). `Biolord(adata, n_latent=32, gene_likelihood=normal)`; `max_epochs=100`, `batch_size=256`. Test-time `model.predict(ctrl_slice)` returns (mean, var); we sample $\hat{X} = \mu + \sigma \epsilon$.

sVAE+ / SAMS-VAE ([Bereket & Karaletsos, 2023](#)). `SAMSVAEModel(n_latent=32, mask_prior_prob=0.01)`; train 30 epochs at `lr=5e-4` with `grad-clip 5.0` (paper default). Test-time: `SAMSVAEPredictor.sample_observations(...)`.

GEARS-uncertainty ([Roohani et al., 2024](#)). `cell-gears 0.1.2` with the official Norman PertData; train at `hidden_size=64`, `batch_size=64` for 5 epochs on Modal A100.

STATE-SE-600M ([Adduri et al., 2025](#)). Inference via `state tx infer -model-dir snapshot/fewshot,zeroshot/cell_line -checkpoint final.ckpt -adata input.h5ad`. Modal image pins `arc-state==0.9.32` and `transformers>=4.52.3,<5.0` (transformers 5.x rejects ST checkpoints’ `hidden_size=328` via `huggingface_hub.dataclasses.validate_architecture`). ST-SE-Tahoe download requires a CPU pre-stage workaround (state_prestage entrypoint with `hf_transfer` enabled): `snapshot_download` inside a GPU function triggers Modal worker heartbeat starvation on the 113-file snapshot, and HF Hub rejects the largest weight file via standard single-stream HTTP.

J. Noise Variants for Point Predictors

The Mean predictor and Bilinear ridge produce per-perturbation mean vectors with no native cell-level variance. To compute distributional discrepancies, ConfPert samples n_{cells} synthetic cells per a documented variant. All

four variants are reported in K1 with no cherry-picking; noise statistics are fit on observed train-fold cells (no test leak).

A No noise: $X_i^{\text{pred}} = \mu_p$. Variance ratio collapses to 0 by construction.

B Isotropic: $X_i^{\text{pred}} = \mu_p + \sigma_{\text{global}}\epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, I_d)$.

C Per-gene marginal: $X_{i,j}^{\text{pred}} = \mu_{p,j} + \sigma_{p,j}\epsilon_{i,j}$.

D Full covariance: $X_i^{\text{pred}} = \mu_p + L\epsilon_i$, where L is Cholesky of $\Sigma_p + 10^{-4}I$ (eigendecomposition fallback if Cholesky fails).

K. Dataset Preprocessing

Perturb-seq. Locate the perturbation column via the regex `ctrl|control|safe|ntc|non-target|non_target|nontarget|scramble`; select top- N perturbations ($N = 50$ within-perturbation, $N = 200$ cross-cell-line); filter to ≥ 30 cells / perturbation; `sc.pp.filter_cells(min_counts=200)`; `sc.pp.filter_genes(min_cells=5)`; `sc.pp.normalize_total(target_sum=1e4)`; `sc.pp.log1p`; `sc.pp.highly_variable_genes(n_top_genes=512)`; subset to HVGs.

Tahoe-100M. Same preprocessing as Perturb-seq plus drop a literal-`NaN` `var_names` entry that breaks `AnnData` indexing in the Tahoe-100M release.

PRISM + Hallmark. PRISM Repurposing Public 24Q2 LFC matrix (905 cell lines $\times$ 1518 drugs after QC-Failure filter and per-(cell, compound) replicate mean) via Figshare file IDs 46630978, 46630981, 46631056, 46631146. Hallmark MSigDB v2024.1.Hs (50 sets, 4384 unique gene symbols).

L. Hyperparameter Summary

Table 5. Predictor training hyperparameters.

Predictor	Backbone / lib	Epochs	Batch	Other
Mean	numpy	—	—	4 noise variants A–D
Bilinear ridge	numpy / scikit-learn	—	—	$K = 10$ PCs, $\lambda = 0.1$
Noisy mean	numpy	—	—	per-gene std calibration
scGen	scgen 2.1.0 + scvi-tools <1.4	100	128	ES patience 10
CPA	cpa-tools 0.8.8 + scvi <1.0	50	256	$n_{\text{lat}} = 64$, recon=gauss
biolord	biolord 0.0.3 + scvi-tools <1.4	100	256	$n_{\text{lat}} = 32$, step_size_lr 45
sVAE+/SAMS-VAE	insitro/sams-vaе (git)	30	128	$n_{\text{lat}} = 32$, lr 5e-4
GEARS-unc	cell-gears 0.1.2	5	64	hidden_size 64
STATE-SE-600M	arc-state 0.9.32 + transformers 4.57.6	—	—	ST-SE-Reprogle / ST-SE-Tahoe ckpts

Conformal-side: $\alpha \in \{0.05, 0.10, 0.20\}$, calibration / test split 50/50 over held-out perturbations, random seed 42.

M. Compute

Lightweight predictors (Mean, Bilinear ridge, Noisy mean) run on a 2024 Apple MacBook Pro (M3 Pro, 36 GB RAM). Heavyweight predictors run on Modal A100-40GB workers. Per-predictor Modal spend: scGen $\sim$ \$0.20 / dataset, biolord $\sim$ \$0.15, CPA $\sim$ \$1.50, sVAE+ $\sim$ \$0.20, GEARS $\sim$ \$3, STATE $\sim$ \$1. Cumulative ConfPert Modal spend $\approx$ \$180.